🇨🇦 Canada Import Gap — Yiwu Small Commo Opportunity Analysis

Mondoré Consulting · Strategic Intelligence Report

Objective: Identify export opportunities in the Canadian market for products supplied by the **Yiw wholesale market** — small consumer commodities including toys, bags, kitchenware, home déc apparel accessories, and seasonal goods.

Methodology:

1. **Macro scan** — compare Canada's total world imports vs. imports from China across all HS 2 chapters
2. **Universe filter** — isolate chapters that constitute the Yiwu small commodity market
3. **Gap analysis** — for each Yiwu segment: Gap = World Imports − China Imports
4. **Opportunity Score** = Gap × (1 − China Share%) — rewards large gap **and** low Chinese penetration
5. **HS 6 deep dive** — drill into 6-digit product codes for highest-score segments

Data source: UN Comtrade API · Canada (reporter 124) · Years 2021–202

In [1]:

```python
# ── Environment setup ────────────────────────────────────────
import sys, os
sys.path.insert(0, os.path.dirname(os.path.abspath('__file__')))

import warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.gridspec as gridspec
from matplotlib.colors import LinearSegmentedColormap
import seaborn as sns
from IPython.display import display, HTML

# Chart style — clean white for PDF export
plt.rcParams.update({
    'figure.facecolor':  'white',
    'axes.facecolor':    '#F8F9FA',
    'axes.edgecolor':    '#DEE2E6',
    'axes.labelcolor':   '#212529',
    'xtick.color':       '#495057',
    'ytick.color':       '#495057',
    'text.color':        '#212529',
    'grid.color':        '#DEE2E6',
    'grid.linestyle':    '--',
    'grid.alpha':        0.7,
```

```python
    'font.family':       'DejaVu Sans',
    'font.size':         10,
    'axes.titlesize':    13,
    'axes.titleweight':  'bold',
    'axes.labelsize':    10,
    'figure.dpi':        130,
})

CANADA_RED   = '#D52B1E'
CHINA_GOLD   = '#F4A300'
GAP_BLUE     = '#1A73E8'
ACCENT_TEAL  = '#00897B'
LIGHT_GRAY   = '#E9ECEF'

print('✅ Libraries loaded')
```

✅ Libraries loaded

In [2]:
```python
# ── Import project modules ──────────────────────────────────────────────
from config import CHINA_CODE, WORLD_CODE, DEFAULT_YEARS
from api import fetch_hs2_all_years, fetch_hs6_chapter
from analysis import records_to_df

YEARS = [2021, 2022, 2023, 2024]

print(f'Loading data for years: {YEARS}')
print('This may take a moment on first run (API calls are cached locally)

# ── Load HS2 data — try live API, fall back to demo ──────────────────
USE_DEMO = False   # set True to use synthetic demo data

try:
    world_r = fetch_hs2_all_years(WORLD_CODE, YEARS, use_demo=USE_DEMO)
    china_r = fetch_hs2_all_years(CHINA_CODE, YEARS, use_demo=USE_DEMO)
    if not world_r:
        raise ValueError('Empty response from API')
    DATA_MODE = '🌐 Live — UN Comtrade API'
except Exception as e:
    print(f'⚠️  Live API unavailable ({e}). Falling back to demo data.')
    world_r = fetch_hs2_all_years(WORLD_CODE, YEARS, use_demo=True)
    china_r = fetch_hs2_all_years(CHINA_CODE, YEARS, use_demo=True)
    DATA_MODE = '📦 Demo — Synthetic (calibrated to Statistics Canada)'

print(f'Data mode: {DATA_MODE}')
print(f'World records: {len(world_r):,}   China records: {len(china_r):,}
```

```
Loading data for years: [2021, 2022, 2023, 2024]
This may take a moment on first run (API calls are cached locally)...
  [api]   2021 partner=0 cmd=AG2 — attempt 1
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=AG2&maxRecords=
&format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to pro
OSError('Tunnel connection failed: 403 Forbidden')))
  [api]   2021 partner=0 cmd=AG2 — attempt 2
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=AG2&maxRecords=
&format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to pro
OSError('Tunnel connection failed: 403 Forbidden')))
```

```
  [api]   2021 partner=0 cmd=AG2 — attempt 3
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=AG2&maxRecords=
&format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to pro
OSError('Tunnel connection failed: 403 Forbidden')))
⚠️  Live API unavailable (HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=AG2&maxRecords=
&format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to pro
OSError('Tunnel connection failed: 403 Forbidden')))). Falling back to demo d
  [demo] Generating synthetic HS2 records for partner=0, years=[2021, 2022, 2
2024]
  [demo] Generating synthetic HS2 records for partner=156, years=[2021, 2022,
2024]
Data mode: 📦 Demo — Synthetic (calibrated to Statistics Canada)
World records: 392   China records: 392
```

In [3]:

```
# ── Build HS2 summary DataFrame ──────────────────────────────────────
world_df = records_to_df(world_r)
china_df = records_to_df(china_r)

world_agg = (
    world_df
    .groupby(['hs_code', 'description'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'world_usd'})
)
china_agg = (
    china_df
    .groupby(['hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'china_usd'})
)

hs2 = world_agg.merge(china_agg, on='hs_code', how='left')
hs2['china_usd']       = hs2['china_usd'].fillna(0)
hs2['gap_usd']         = hs2['world_usd'] - hs2['china_usd']
hs2['china_share_pct'] = (hs2['china_usd'] / hs2['world_usd'].replace(0,
hs2['world_b']         = (hs2['world_usd'] / 1e9).round(3)
hs2['china_b']         = (hs2['china_usd'] / 1e9).round(3)
hs2['gap_b']           = (hs2['gap_usd'] / 1e9).round(3)
hs2['opp_score']       = (hs2['gap_usd'] * (1 - hs2['china_share_pct'] /
hs2['short_desc']      = hs2['description'].str[:40]
hs2['label']           = hs2['hs_code'] + ' · ' + hs2['short_desc']
hs2 = hs2.sort_values('opp_score', ascending=False).reset_index(drop=True
hs2.index += 1
hs2.index.name = 'rank'

print(f'HS2 chapters loaded: {len(hs2)}')
print(f'Total Canada imports (all years): USD {hs2["world_b"].sum():,.0f}
print(f'China share (average): {(hs2["china_b"].sum() / hs2["world_b"].su
```

```
HS2 chapters loaded: 98
Total Canada imports (all years): USD 1,801 B
China share (average): 19.8%
```

Section 1 — Macro Overview: All Canadian Impor

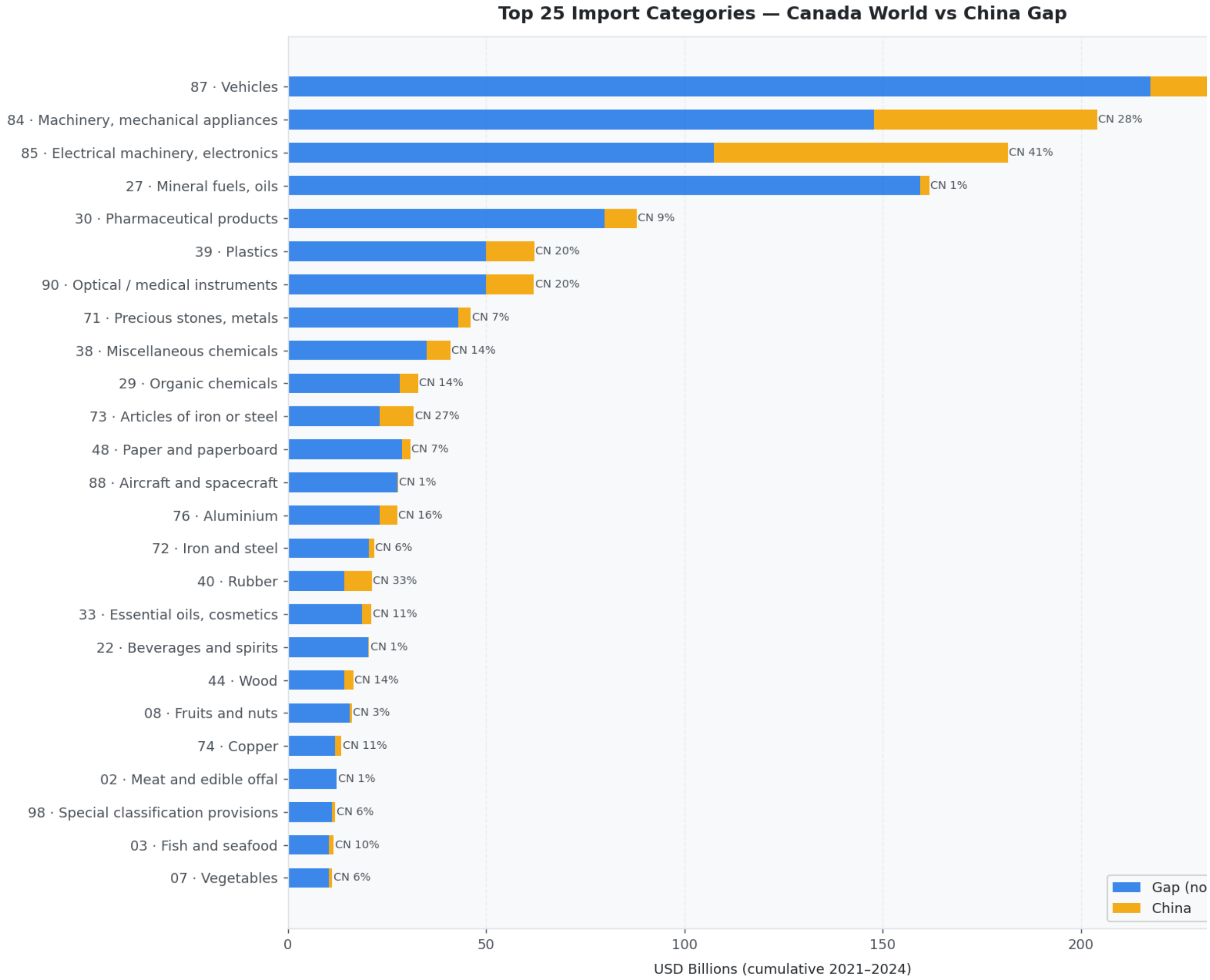
Before zooming into Yiwu categories, we scan the entire import landscape to understand the siz
structure of Canada's trade deficit with China across all sectors.

In [4]:
```
# ── KPI Summary ──────────────────────────────────────────────────────────────
total_world = hs2['world_b'].sum()
total_china = hs2['china_b'].sum()
total_gap   = hs2['gap_b'].sum()
avg_share   = total_china / total_world * 100
n_low       = (hs2['china_share_pct'] < 15).sum()

fig, axes = plt.subplots(1, 5, figsize=(17, 2.5))
fig.suptitle(f'Canada Imports — All HS2 Chapters · {YEARS[0]}–{YEARS[-1]}
             fontsize=11, y=1.02, color='#495057')

kpis = [
    ('Total World Imports', f'USD {total_world:,.0f} B',  CANADA_RED),
    ('Total China Imports', f'USD {total_china:,.0f} B',  CHINA_GOLD),
    ('Total Import Gap',    f'USD {total_gap:,.0f} B',    GAP_BLUE),
    ('China Avg Share',     f'{avg_share:.1f}%',           ACCENT_TEAL),
    ('Categories CN<15%',   f'{n_low} chapters',          '#7B1FA2'),
]
for ax, (title, val, color) in zip(axes, kpis):
    ax.set_facecolor(color)
    ax.text(0.5, 0.65, val,    ha='center', va='center', fontsize=18, fon
    ax.text(0.5, 0.25, title,  ha='center', va='center', fontsize=9,  col
    ax.set_xticks([]); ax.set_yticks([])
    for spine in ax.spines.values(): spine.set_visible(False)

plt.tight_layout()
plt.savefig('fig_01_kpis.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 1 saved.')
```


```
Figure 1 saved.
```

In [5]:
```
# ── Figure 2: Top 25 Gaps — All HS2 ─────────────────────────────────────────
top25 = hs2.head(25).sort_values('world_b', ascending=True)

fig, ax = plt.subplots(figsize=(13, 10))

bars_gap   = ax.barh(top25['label'], top25['gap_b'],   color=GAP_BLUE,
bars_china = ax.barh(top25['label'], top25['china_b'], color=CHINA_GOLD,
                     left=top25['gap_b'])

for bar, row in zip(bars_gap, top25.itertuples()):
    ax.text(bar.get_width() + row.china_b + 0.3, bar.get_y() + bar.get_he
            f'CN {row.china_share_pct:.0f}%', va='center', fontsize=8, co

ax.set_xlabel('USD Billions (cumulative 2021–2024)', labelpad=8)
```

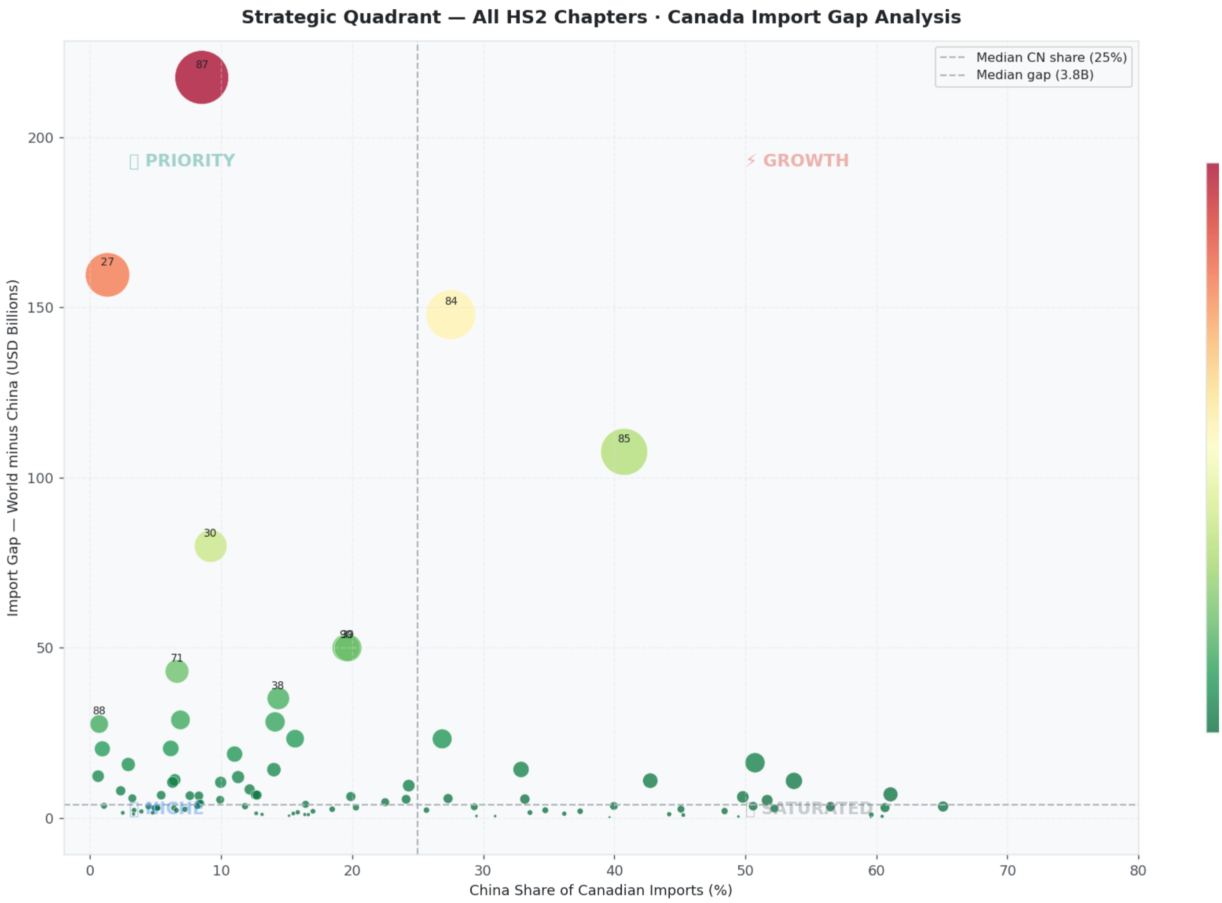
```
ax.set_title('Top 25 Import Categories — Canada World vs China Gap', pad=
ax.legend(loc='lower right', framealpha=0.9)
ax.grid(axis='x', alpha=0.5)
ax.set_axisbelow(True)
plt.tight_layout()
plt.savefig('fig_02_top25_gaps.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 2 saved.')
```




```
Figure 2 saved.
```

In [6]:
```
# ── Figure 3: Strategic Quadrant — All HS2 ──────────────────────────────
fig, ax = plt.subplots(figsize=(14, 9))

med_share = 25
med_gap   = hs2['gap_b'].median()

scatter = ax.scatter(
    hs2['china_share_pct'], hs2['gap_b'],
    s=hs2['world_b'] * 6,
    c=hs2['opp_score'], cmap='RdYlGn_r',
    alpha=0.75, edgecolors='white', linewidth=0.4,
)

plt.colorbar(scatter, ax=ax, label='Opportunity Score (B USD)', shrink=0.

ax.axvline(med_share, color='#ADB5BD', linestyle='--', lw=1.2, label=f'Me
ax.axhline(med_gap,   color='#ADB5BD', linestyle='--', lw=1.2, label=f'Me

# Quadrant labels
y_max = hs2['gap_b'].max()
```

```
quad_style = dict(fontsize=12, fontweight='bold', alpha=0.35)
ax.text(3,   y_max * 0.88, '🎯 PRIORITY',   color=ACCENT_TEAL, **quad_sty
ax.text(50,  y_max * 0.88, '⚡ GROWTH',      color=CANADA_RED,  **quad_st
ax.text(3,   med_gap * 0.3, '🔍 NICHE',      color=GAP_BLUE,    **quad_st
ax.text(50,  med_gap * 0.3, '🔒 SATURATED',  color='#6C757D',   **quad_st

# Annotate top 10 by score
for _, row in hs2.head(10).iterrows():
    ax.annotate(row['hs_code'], (row['china_share_pct'], row['gap_b']),
                fontsize=7.5, ha='center', va='bottom',
                xytext=(0, 5), textcoords='offset points', color='#212529

ax.set_xlabel('China Share of Canadian Imports (%)')
ax.set_ylabel('Import Gap — World minus China (USD Billions)')
ax.set_title('Strategic Quadrant — All HS2 Chapters · Canada Import Gap A
ax.legend(fontsize=9, loc='upper right')
ax.set_xlim(-2, 80)
ax.grid(True, alpha=0.4)
plt.tight_layout()
plt.savefig('fig_03_quadrant_all.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 3 saved.')
```




```
Figure 3 saved.
```

Section 2 — Yiwu Small Commodity Universe

The **Yiwu wholesale market** (义乌) is the world's largest small commodity trading hub, supplyin 1.8 million product types across 75,000+ booths. Its core export strengths align w following HS2 chapters:

Segment	HS Chapters	Products
Bags & Travel Goods	42	Handbags, backpacks, wallets, luggage
Kitchenware & Tableware	39, 69, 70, 73, 82	Plastic containers, ceramics, glassware, steel cutlery
Home Décor & Hardware	83, 94	Frames, hooks, locks, lamps, c furniture
Textiles & Apparel	57, 61, 62, 63, 64, 65, 66	Carpets, clothing, towels, shoe umbrellas
Toys, Games & Sports	95	Toys, puzzles, outdoor sports equipment
Seasonal & Festive	46, 67, 96	Artificial flowers, wicker, statior zippers

In [7]:

```
# ── Yiwu Universe Definition ──────────────────────────────────────
YIWU_SEGMENTS = {
    'Bags & Travel Goods':         ['42'],
    'Kitchenware & Tableware':     ['39', '69', '70', '73', '82'],
    'Home Décor & Hardware':       ['83', '94'],
    'Textiles & Apparel':          ['57', '61', '62', '63', '64', '65', '6
    'Toys, Games & Sports':        ['95'],
    'Seasonal & Festive Goods':    ['46', '67', '96'],
}

# Create chapter → segment map
chapter_to_segment = {}
for seg, chapters in YIWU_SEGMENTS.items():
    for ch in chapters:
        chapter_to_segment[ch.zfill(2)] = seg

YIWU_CHAPTERS = list(chapter_to_segment.keys())

# Filter HS2 to Yiwu universe
yiwu = hs2[hs2['hs_code'].isin(YIWU_CHAPTERS)].copy()
yiwu['segment'] = yiwu['hs_code'].map(chapter_to_segment)
yiwu = yiwu.sort_values('opp_score', ascending=False).reset_index(drop=Tr
yiwu.index += 1
yiwu.index.name = 'rank'

print(f'Yiwu universe: {len(yiwu)} HS2 chapters across {len(YIWU_SEGMENTS
print(f'Total Yiwu-relevant world imports: USD {yiwu["world_b"].sum():,.1
print(f'China currently captures:          USD {yiwu["china_b"].sum():,.1
print(f'Remaining gap (opportunity):       USD {yiwu["gap_b"].sum():,.1f}
print(f'Average China share in these categories: {(yiwu["china_b"].sum()
```

```
Yiwu universe: 19 HS2 chapters across 6 segments
Total Yiwu-relevant world imports: USD 262.9 B
China currently captures:          USD 107.5 B
Remaining gap (opportunity):       USD 155.4 B
Average China share in these categories: 40.9%
```

In [8]:

```
# ── Figure 4: Yiwu Universe — Segment Summary ─────────────────────
seg_summary = (
    yiwu.groupby('segment', as_index=False)
    .agg(world_b=('world_b','sum'), china_b=('china_b','sum'),
         gap_b=('gap_b','sum'),     opp_score=('opp_score','sum'))
```

```
)
seg_summary['share_pct'] = (seg_summary['china_b'] / seg_summary['world_b
seg_summary = seg_summary.sort_values('opp_score', ascending=False).reset

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Left: stacked bar by segment
ax = axes[0]
segs = seg_summary['segment']
x = np.arange(len(segs))
b1 = ax.bar(x, seg_summary['gap_b'],   color=GAP_BLUE,   label='Gap (non-
b2 = ax.bar(x, seg_summary['china_b'], color=CHINA_GOLD, label='China sha
            bottom=seg_summary['gap_b'])
for i, (g, c, s) in enumerate(zip(seg_summary['gap_b'], seg_summary['chin
    ax.text(i, g + c + 0.3, f'CN {s:.0f}%', ha='center', va='bottom', fon
ax.set_xticks(x)
ax.set_xticklabels([s.replace(' & ', '\n& ') for s in segs], fontsize=8.5
ax.set_ylabel('USD Billions (2021–2024 cumulative)')
ax.set_title('Yiwu Segments — World Import Volume', pad=10)
ax.legend()
ax.grid(axis='y', alpha=0.5)
ax.set_axisbelow(True)

# Right: opportunity score pie
ax2 = axes[1]
colors_pie = [CANADA_RED, CHINA_GOLD, GAP_BLUE, ACCENT_TEAL, '#7B1FA2', '
wedges, texts, autotexts = ax2.pie(
    seg_summary['opp_score'],
    labels=seg_summary['segment'],
    colors=colors_pie,
    autopct='%1.1f%%',
    startangle=120,
    pctdistance=0.75,
    labeldistance=1.08,
    textprops={'fontsize': 8.5},
    wedgeprops={'linewidth': 1.5, 'edgecolor': 'white'},
)
for at in autotexts: at.set_fontsize(8)
ax2.set_title('Share of Total Opportunity Score\nby Yiwu Segment', pad=10

plt.suptitle('Yiwu Small Commodity Universe — Canada Import Opportunity',
plt.tight_layout()
plt.savefig('fig_04_yiwu_segments.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 4 saved.')
```

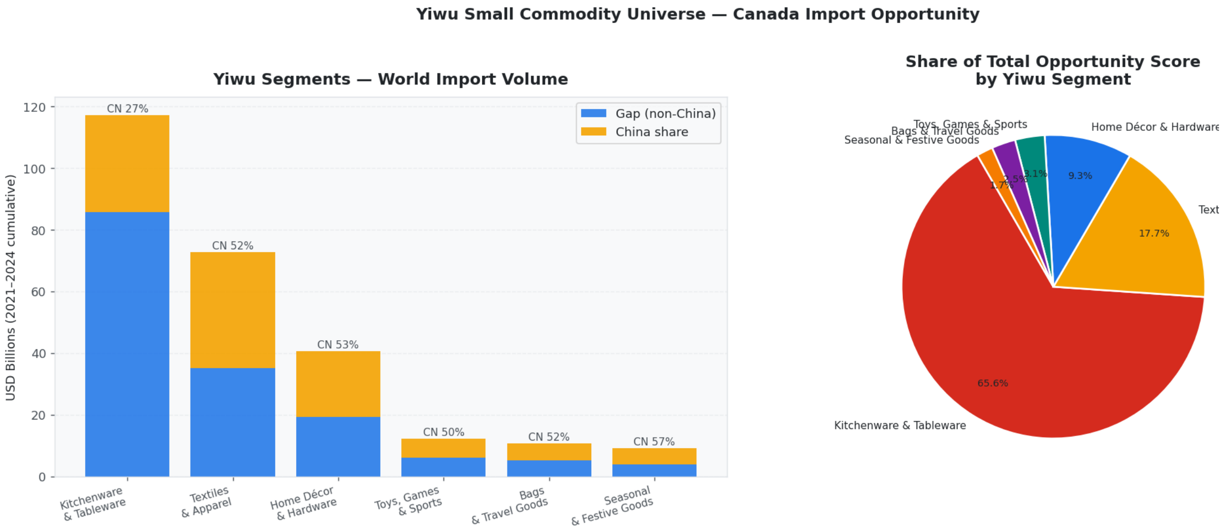
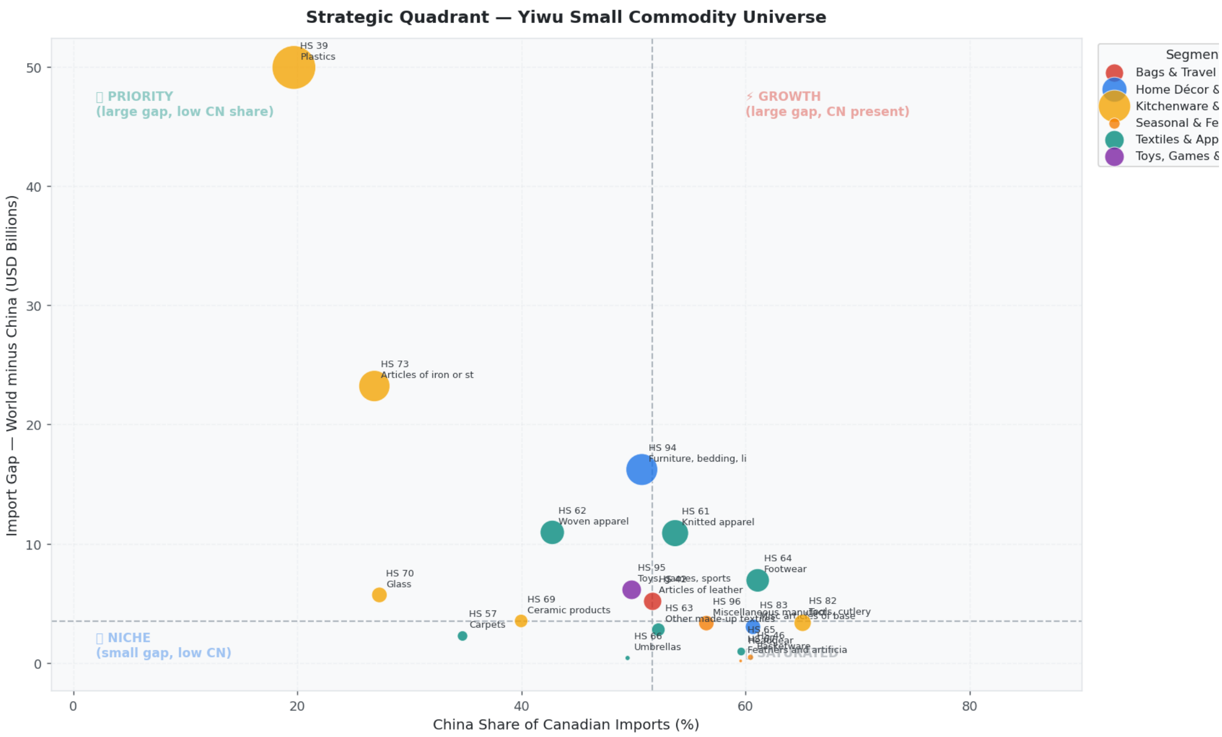


Figure 4 saved.

In [9]:
```python
# ── Figure 5: Yiwu Strategic Quadrant ──────────────────────────────────
SEGMENT_COLORS = {
    'Bags & Travel Goods':         CANADA_RED,
    'Kitchenware & Tableware':     CHINA_GOLD,
    'Home Décor & Hardware':       GAP_BLUE,
    'Textiles & Apparel':          ACCENT_TEAL,
    'Toys, Games & Sports':        '#7B1FA2',
    'Seasonal & Festive Goods':    '#F57C00',
}

fig, ax = plt.subplots(figsize=(14, 8))

for seg, grp in yiwu.groupby('segment'):
    ax.scatter(
        grp['china_share_pct'], grp['gap_b'],
        s=grp['world_b'] * 18,
        c=SEGMENT_COLORS.get(seg, '#999'),
        alpha=0.78, edgecolors='white', linewidth=0.6,
        label=seg, zorder=3,
    )
    for _, row in grp.iterrows():
        ax.annotate(
            f"HS {row['hs_code']}\n{row['description'][:22]}",
            (row['china_share_pct'], row['gap_b']),
            fontsize=7.2, ha='left', va='bottom',
            xytext=(5, 5), textcoords='offset points', color='#343A40',
        )

med_share_y = yiwu['china_share_pct'].median()
med_gap_y   = yiwu['gap_b'].median()
ax.axvline(med_share_y, color='#ADB5BD', linestyle='--', lw=1.2)
ax.axhline(med_gap_y,   color='#ADB5BD', linestyle='--', lw=1.2)

y_max2 = yiwu['gap_b'].max()
ax.text(2,  y_max2 * 0.92, '🎯 PRIORITY\n(large gap, low CN share)',  for
ax.text(60, y_max2 * 0.92, '⚡ GROWTH\n(large gap, CN present)',        fc
ax.text(2,  med_gap_y*0.15, '🔍 NICHE\n(small gap, low CN)',            fc
ax.text(60, med_gap_y*0.15, '🔒 SATURATED',                              f

ax.set_xlabel('China Share of Canadian Imports (%)', fontsize=11)
ax.set_ylabel('Import Gap — World minus China (USD Billions)', fontsize=1
ax.set_title('Strategic Quadrant — Yiwu Small Commodity Universe', pad=12
```

```
ax.legend(title='Segment', bbox_to_anchor=(1.01, 1), loc='upper left', fo
ax.set_xlim(-2, 90)
ax.grid(True, alpha=0.35)
plt.tight_layout()
plt.savefig('fig_05_yiwu_quadrant.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 5 saved.')
```




```
Figure 5 saved.
```

Section 3 — Segment Deep Dive

For each of the 6 Yiwu segments, we show: market size, China share, gap size, and oppo score per HS 2 chapter.

In [10]:
```
# ── Figure 6: Per-Segment Breakdown — 6-panel grid ──────────────────
segments_ordered = seg_summary['segment'].tolist()
n_seg = len(segments_ordered)
ncols = 3
nrows = (n_seg + ncols - 1) // ncols

fig, axes = plt.subplots(nrows, ncols, figsize=(18, nrows * 5.5))
axes_flat = axes.flatten()

for idx, seg in enumerate(segments_ordered):
    ax = axes_flat[idx]
    grp = yiwu[yiwu['segment'] == seg].sort_values('gap_b', ascending=Tru
    color = SEGMENT_COLORS.get(seg, '#888')

    b_gap   = ax.barh(grp['label'], grp['gap_b'],   color=color,      alp
    b_china = ax.barh(grp['label'], grp['china_b'], color=CHINA_GOLD, alp
                      left=grp['gap_b'])

    for bar, row in zip(b_gap, grp.itertuples()):
        total = row.gap_b + row.china_b
```

```
        ax.text(total + 0.05, bar.get_y() + bar.get_height()/2,
                f'CN {row.china_share_pct:.0f}% · Score {row.opp_score:.2
                va='center', fontsize=7.5, color='#495057')

    ax.set_title(seg, pad=8, color=color, fontsize=11)
    ax.set_xlabel('USD Billions')
    ax.legend(fontsize=8, loc='lower right')
    ax.grid(axis='x', alpha=0.4)
    ax.set_axisbelow(True)

# hide empty panels
for j in range(n_seg, len(axes_flat)):
    axes_flat[j].set_visible(False)

plt.suptitle('Yiwu Segments — HS2 Chapter Detail (Gap vs China Share)', f
plt.tight_layout()
plt.savefig('fig_06_segment_detail.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 6 saved.')
```


```
Figure 6 saved.
```

Section 4 — HS 6 Deep Dive: Top 3 Opportunity Chapters

We drill into 6 -digit product codes for the top 3 HS 2 chapters by opportunity score t specific products with the highest potential.

In [11]:
```
# ── Load HS6 data for top 3 Yiwu chapters ──────────────────────────
top3_chapters = yiwu.head(3)['hs_code'].tolist()
print(f'Loading HS6 data for chapters: {top3_chapters}')
```

```python
hs6_data = {}
for ch in top3_chapters:
    try:
        w6 = fetch_hs6_chapter(WORLD_CODE, ch, YEARS, use_demo=USE_DEMO)
        c6 = fetch_hs6_chapter(CHINA_CODE, ch, YEARS, use_demo=USE_DEMO)
        if not w6: raise ValueError('Empty')
    except Exception as e:
        print(f'  ⚠️  HS6 live API failed for ch {ch}: {e} — using demo')
        w6 = fetch_hs6_chapter(WORLD_CODE, ch, YEARS, use_demo=True)
        c6 = fetch_hs6_chapter(CHINA_CODE, ch, YEARS, use_demo=True)

    w6df = records_to_df(w6)
    c6df = records_to_df(c6)

    w6a = w6df.groupby(['hs_code','description'], as_index=False)['value_
    c6a = c6df.groupby(['hs_code'], as_index=False)['value_usd'].sum().re

    m6 = w6a.merge(c6a, on='hs_code', how='left')
    m6['china_usd']  = m6['china_usd'].fillna(0)
    m6['gap_usd']    = m6['world_usd'] - m6['china_usd']
    m6['share_pct']  = (m6['china_usd'] / m6['world_usd'].replace(0,np.na
    m6['world_m']    = (m6['world_usd']/1e6).round(2)
    m6['china_m']    = (m6['china_usd']/1e6).round(2)
    m6['gap_m']      = (m6['gap_usd']/1e6).round(2)
    m6['opp_score']  = (m6['gap_usd'] * (1 - m6['share_pct']/100) / 1e9).
    m6 = m6[m6['world_m'] > 0].sort_values('opp_score', ascending=False).
    m6.index += 1; m6.index.name = 'rank'
    hs6_data[ch] = m6
    print(f'  Chapter {ch}: {len(m6)} products loaded')
```

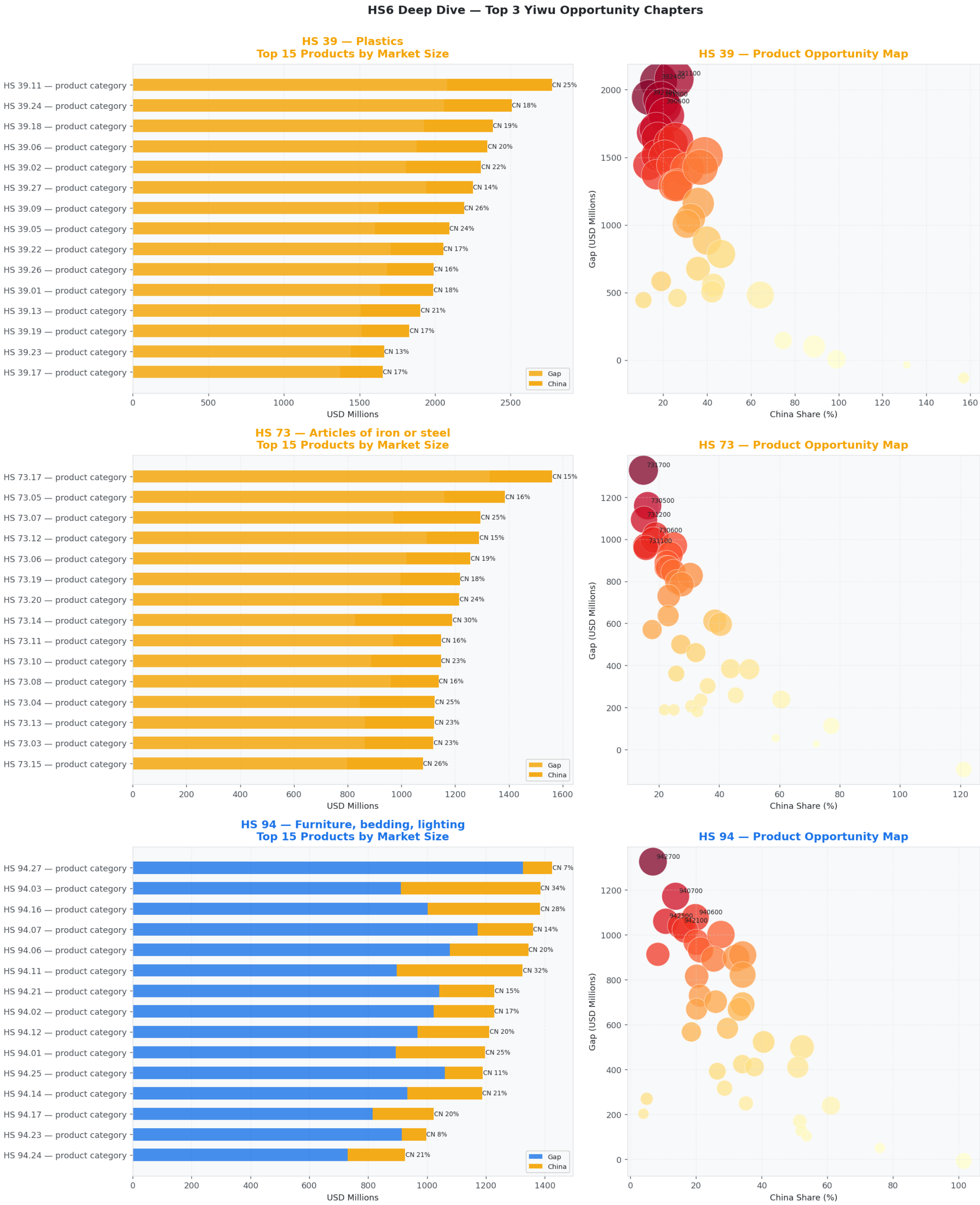
```
Loading HS6 data for chapters: ['39', '73', '94']
  [api]   2021 partner=0 cmd=39 — attempt 1
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=39&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  [api]   2021 partner=0 cmd=39 — attempt 2
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=39&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
```

```
  [api]   2021 partner=0 cmd=39 — attempt 3
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=39&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  ⚠️  HS6 live API failed for ch 39: HTTPSConnectionPool(host='comtradeapi.un
port=443): Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=39&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden'))) — using demo
  [demo] Generating synthetic HS6 records for chapter=39, partner=0, years=[2
2022, 2023, 2024]
  [demo] Generating synthetic HS6 records for chapter=39, partner=156, years=
2022, 2023, 2024]
  Chapter 39: 39 products loaded
  [api]   2021 partner=0 cmd=73 — attempt 1
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=73&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  [api]   2021 partner=0 cmd=73 — attempt 2
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=73&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  [api]   2021 partner=0 cmd=73 — attempt 3
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=73&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  ⚠️  HS6 live API failed for ch 73: HTTPSConnectionPool(host='comtradeapi.un
port=443): Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=73&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden'))) — using demo
  [demo] Generating synthetic HS6 records for chapter=73, partner=0, years=[2
2022, 2023, 2024]
  [demo] Generating synthetic HS6 records for chapter=73, partner=156, years=
2022, 2023, 2024]
  Chapter 73: 38 products loaded
  [api]   2021 partner=0 cmd=94 — attempt 1
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=94&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  [api]   2021 partner=0 cmd=94 — attempt 2
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=94&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
```

```
  [api]   2021 partner=0 cmd=94 — attempt 3
  [error] Request failed: HTTPSConnectionPool(host='comtradeapi.un.org', port
Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=94&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden')))
  ⚠️  HS6 live API failed for ch 94: HTTPSConnectionPool(host='comtradeapi.un
port=443): Max retries exceeded with url: /data/v1/get/C/A/HS?
reporterCode=124&partnerCode=0&period=2021&flowCode=M&cmdCode=94&maxRecords=1
format=JSON&includeDesc=true (Caused by ProxyError('Unable to connect to prox
OSError('Tunnel connection failed: 403 Forbidden'))) — using demo
  [demo] Generating synthetic HS6 records for chapter=94, partner=0, years=[2
2022, 2023, 2024]
  [demo] Generating synthetic HS6 records for chapter=94, partner=156, years=
2022, 2023, 2024]
  Chapter 94: 38 products loaded
```

In [12]:
```
# —— Figure 7: HS6 Deep Dive — Top 3 Chapters ——————————————————————————
fig, axes = plt.subplots(3, 2, figsize=(18, 20))

for row_idx, ch in enumerate(top3_chapters):
    df6 = hs6_data[ch]
    ch_name = yiwu[yiwu['hs_code'] == ch]['description'].iloc[0]
    color = SEGMENT_COLORS.get(yiwu[yiwu['hs_code'] == ch]['segment'].ilo

    top15 = df6.head(15).sort_values('world_m', ascending=True)

    # Left panel: bar chart
    ax_bar = axes[row_idx][0]
    ax_bar.barh(top15['description'].str[:45], top15['gap_m'],   color=co
    ax_bar.barh(top15['description'].str[:45], top15['china_m'], color=CH
                left=top15['gap_m'])
    for bar, (_, r) in zip(ax_bar.patches[:len(top15)], top15.iterrows())
        total_m = r['gap_m'] + r['china_m']
        ax_bar.text(total_m + 1, bar.get_y() + bar.get_height()/2,
                    f"CN {r['share_pct']:.0f}%", va='center', fontsize=7.
    ax_bar.set_title(f'HS {ch} — {ch_name[:50]}\nTop 15 Products by Marke
    ax_bar.set_xlabel('USD Millions')
    ax_bar.legend(fontsize=8)
    ax_bar.grid(axis='x', alpha=0.4); ax_bar.set_axisbelow(True)

    # Right panel: scatter opportunity
    ax_sc = axes[row_idx][1]
    sc = ax_sc.scatter(
        df6['share_pct'], df6['gap_m'],
        s=df6['world_m'] * 0.8,
        c=df6['opp_score'], cmap='YlOrRd',
        alpha=0.75, edgecolors='white', linewidth=0.5,
    )
    plt.colorbar(sc, ax=ax_sc, label='Opp Score (B)', shrink=0.8)
    # label top 5
    for _, r in df6.head(5).iterrows():
        ax_sc.annotate(r['hs_code'], (r['share_pct'], r['gap_m']),
                       fontsize=7.5, xytext=(4,4), textcoords='offset poi
    ax_sc.set_xlabel('China Share (%)')
    ax_sc.set_ylabel('Gap (USD Millions)')
    ax_sc.set_title(f'HS {ch} — Product Opportunity Map', color=color, pa
    ax_sc.grid(True, alpha=0.35)

plt.suptitle('HS6 Deep Dive — Top 3 Yiwu Opportunity Chapters', fontsize=
```

```
plt.tight_layout()
plt.savefig('fig_07_hs6_deepdive.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 7 saved.')
```



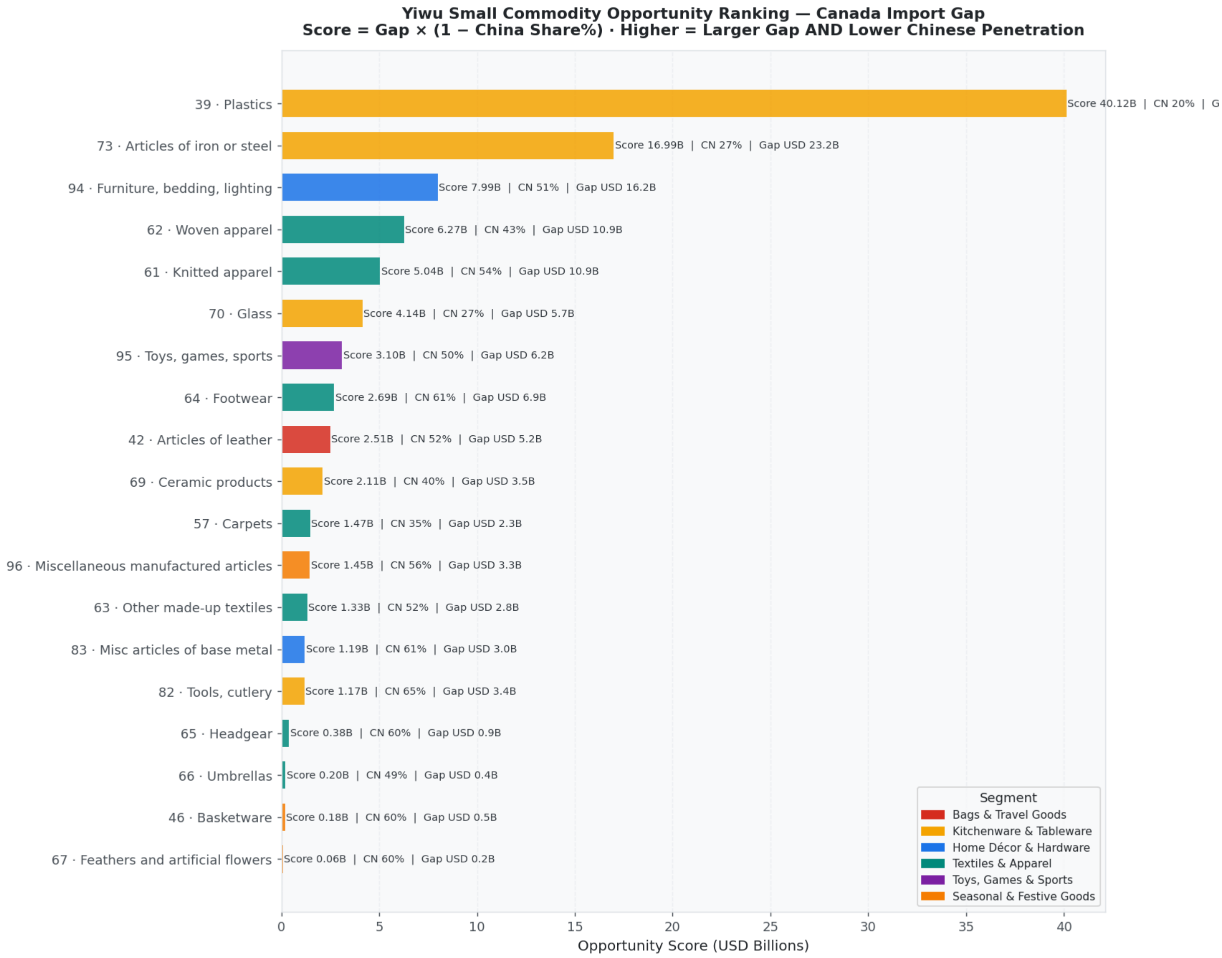

```
Figure 7 saved.
```

Section 5 — Final Opportunity Ranking: Yiwu Sma Commodities

In [13]:

```
# ── Figure 8: Opportunity Score Ranking — Yiwu Only ──────────
yiwu_sorted = yiwu.sort_values('opp_score', ascending=True)
```

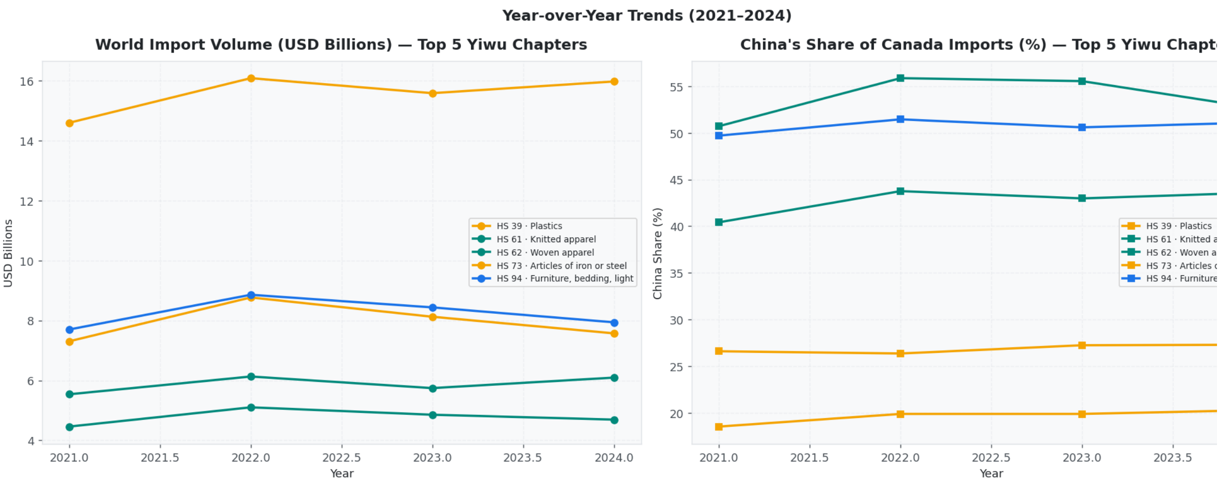
```python
fig, ax = plt.subplots(figsize=(14, max(6, len(yiwu) * 0.55)))

bar_colors = [SEGMENT_COLORS.get(s, '#888') for s in yiwu_sorted['segment
bars = ax.barh(yiwu_sorted['label'], yiwu_sorted['opp_score'], color=bar_

for bar, (_, row) in zip(bars, yiwu_sorted.iterrows()):
    ax.text(bar.get_width() + 0.05,
            bar.get_y() + bar.get_height()/2,
            f"Score {row['opp_score']:.2f}B  |  CN {row['china_share_pct'
            va='center', fontsize=8, color='#343A40')

# Legend
legend_patches = [mpatches.Patch(color=c, label=s) for s, c in SEGMENT_CO
ax.legend(handles=legend_patches, title='Segment', loc='lower right', fon

ax.set_xlabel('Opportunity Score (USD Billions)', fontsize=11)
ax.set_title('Yiwu Small Commodity Opportunity Ranking — Canada Import Ga
             'Score = Gap × (1 − China Share%) · Higher = Larger Gap AND
             pad=12, fontsize=12)
ax.grid(axis='x', alpha=0.4)
ax.set_axisbelow(True)
plt.tight_layout()
plt.savefig('fig_08_yiwu_ranking.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 8 saved.')
```




```
Figure 8 saved.
```

In [14]:
```python
# ── Figure 9: Year-over-Year Trend – Top 5 Yiwu Chapters ──────────────
world_yr = (
    world_df
    .groupby(['year', 'hs_code'], as_index=False)['value_usd']
```

```
    .sum().rename(columns={'value_usd': 'world_usd'})
)
china_yr = (
    china_df
    .groupby(['year', 'hs_code'], as_index=False)['value_usd']
    .sum().rename(columns={'value_usd': 'china_usd'})
)
trend = world_yr.merge(china_yr, on=['year','hs_code'], how='left')
trend['china_usd']  = trend['china_usd'].fillna(0)
trend['share_pct']  = (trend['china_usd'] / trend['world_usd'].replace(0,
trend['gap_m']      = (trend['world_usd'] - trend['china_usd']) / 1e6
trend['world_b']    = trend['world_usd'] / 1e9
trend['year']       = trend['year'].astype(int)

top5_ch = yiwu.head(5)['hs_code'].tolist()
trend5  = trend[trend['hs_code'].isin(top5_ch)].copy()
trend5  = trend5.merge(yiwu[['hs_code','description','segment']].reset_in
trend5['label'] = 'HS ' + trend5['hs_code'] + ' · ' + trend5['description

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

for lab, grp in trend5.groupby('label'):
    ch_code = grp['hs_code'].iloc[0]
    seg     = grp['segment'].iloc[0]
    color   = SEGMENT_COLORS.get(seg, '#888')
    grp_s = grp.sort_values('year')
    ax1.plot(grp_s['year'], grp_s['world_b'], marker='o', color=color, li
    ax2.plot(grp_s['year'], grp_s['share_pct'], marker='s', color=color,

ax1.set_title('World Import Volume (USD Billions) — Top 5 Yiwu Chapters',
ax1.set_xlabel('Year'); ax1.set_ylabel('USD Billions')
ax1.legend(fontsize=7.5); ax1.grid(True, alpha=0.4)

ax2.set_title("China's Share of Canada Imports (%) — Top 5 Yiwu Chapters"
ax2.set_xlabel('Year'); ax2.set_ylabel('China Share (%)')
ax2.legend(fontsize=7.5); ax2.grid(True, alpha=0.4)

plt.suptitle('Year-over-Year Trends (2021–2024)', fontsize=13, fontweight
plt.tight_layout()
plt.savefig('fig_09_trends.png', bbox_inches='tight', dpi=150)
plt.show()
print('Figure 9 saved.')
```




```
Figure 9 saved.
```

In [15]:

```python
# ── Final Ranked Table ──────────────────────────────────────────
final_table = (
    yiwu.reset_index()[['rank','hs_code','description','segment',
                         'world_b','china_b','gap_b','china_share_pct','o
    .copy()
)
final_table.columns = ['Rank','HS2','Description','Segment',
                        'World(B)','China(B)','Gap(B)','CN Share%','Score

print('=== FINAL YIWU OPPORTUNITY RANKING — CANADA IMPORT GAP ===')
print(final_table.to_string(index=False))

# Save to Excel for distribution
final_table.to_excel('yiwu_canada_opportunity.xlsx', index=False, sheet_n
print('\n✅ Saved: yiwu_canada_opportunity.xlsx')
```

```
=== FINAL YIWU OPPORTUNITY RANKING — CANADA IMPORT GAP ===
 Rank HS2                          Description                  Segment  World
China(B)  Gap(B)  CN Share%  Score(B)
    1  39                             Plastics  Kitchenware & Tableware    62.
12.247  49.961       19.69   40.1234
    2  73              Articles of iron or steel  Kitchenware & Tableware
31.764     8.536  23.227      26.87   16.9862
    3  94           Furniture, bedding, lighting    Home Décor & Hardware    32.
16.708  16.218       50.74    7.9892
    4  62                        Woven apparel       Textiles & Apparel
19.115     8.171  10.944      42.75    6.2655
    5  61                      Knitted apparel       Textiles & Apparel    23.
12.624  10.882       53.71    5.0371
    6  70                                Glass  Kitchenware & Tableware
7.838     2.141   5.696      27.32    4.1400
    7  95                  Toys, games, sports     Toys, Games & Sports
12.288     6.120   6.167      49.81    3.0954
    8  64                             Footwear       Textiles & Apparel    17.
10.858   6.920       61.08    2.6931
    9  42                  Articles of leather      Bags & Travel Goods
10.725     5.541   5.184      51.67    2.5053
   10  69                     Ceramic products  Kitchenware & Tableware
5.850     2.338   3.512      39.97    2.1080
   11  57                              Carpets       Textiles & Apparel
3.452     1.199   2.252      34.74    1.4699
   12  96 Miscellaneous manufactured articles Seasonal & Festive Goods
7.657     4.326   3.331      56.50    1.4490
   13  63               Other made-up textiles       Textiles & Apparel
5.822     3.040   2.782      52.22    1.3292
   14  83            Misc articles of base metal    Home Décor & Hardware
7.679     4.658   3.020      60.66    1.1883
   15  82                       Tools, cutlery  Kitchenware & Tableware
9.612     6.257   3.355      65.09    1.1713
   16  65                             Headgear       Textiles & Apparel
2.341     1.395   0.946      59.61    0.3819
   17  66                            Umbrellas       Textiles & Apparel
0.800     0.396   0.404      49.47    0.2042
   18  46                           Basketware Seasonal & Festive Goods
1.172     0.708   0.464      60.44    0.1834
   19  67      Feathers and artificial flowers Seasonal & Festive Goods
0.392     0.234   0.159      59.56    0.0642

✅ Saved: yiwu_canada_opportunity.xlsx
```

In [16]:

```python
# ── Styled display for notebook ──────────────────────────────────────────────
def highlight_score(val):
    if isinstance(val, float):
        if val >= final_table['Score(B)'].quantile(0.75):
            return 'background-color: #d4edda; font-weight: bold'
        elif val >= final_table['Score(B)'].quantile(0.5):
            return 'background-color: #fff3cd'
    return ''

def highlight_segment(val):
    seg_colors_css = {
        'Toys, Games & Sports':      '#e8d5f5',
        'Home Décor & Hardware':     '#d5e8f5',
        'Kitchenware & Tableware':   '#ffeeba',
        'Bags & Travel Goods':       '#f8d7da',
        'Textiles & Apparel':        '#d4edda',
        'Seasonal & Festive Goods':  '#d1ecf1',
    }
    return f"background-color: {seg_colors_css.get(val, 'white')}"

styled = (
    final_table.style
    .applymap(highlight_score, subset=['Score(B)'])
    .applymap(highlight_segment, subset=['Segment'])
    .format({'World(B)': '{:.2f}', 'China(B)': '{:.2f}', 'Gap(B)': '{:.2f
             'CN Share%': '{:.1f}%', 'Score(B)': '{:.3f}'})
    .set_caption('Yiwu Small Commodity Opportunity Ranking — Canada Impor
    .set_table_styles([{
        'selector': 'caption',
        'props': [('font-size', '13px'), ('font-weight', 'bold'), ('color
    }])
)
display(styled)
```

Yiwu Small Commodity Opportunity Ranking — Canada Import Gap Analysis

	Rank	HS 2	Description	Segment	World(B)	China(B)	Gap(B)	Shar
0	1	39	Plastics	Kitchenware & Tableware	62.21	12.25	49.96	19.
1	2	73	Articles of iron or steel	Kitchenware & Tableware	31.76	8.54	23.23	26.
2	3	94	Furniture, bedding, lighting	Home Décor & Hardware	32.93	16.71	16.22	50.
3	4	62	Woven apparel	Textiles & Apparel	19.11	8.17	10.94	42.
4	5	61	Knitted apparel	Textiles & Apparel	23.51	12.62	10.88	53.
5	6	70	Glass	Kitchenware & Tableware	7.84	2.14	5.70	27.
6	7	95	Toys, games, sports	Toys, Games & Sports	12.29	6.12	6.17	49.
7	8	64	Footwear	Textiles & Apparel	17.78	10.86	6.92	61.
8	9	42	Articles of leather	Bags & Travel Goods	10.72	5.54	5.18	51.
9	10	69	Ceramic products	Kitchenware & Tableware	5.85	2.34	3.51	40.
10	11	57	Carpets	Textiles & Apparel	3.45	1.20	2.25	34.
11	12	96	Miscellaneous manufactured articles	Seasonal & Festive Goods	7.66	4.33	3.33	56.
12	13	63	Other made-up textiles	Textiles & Apparel	5.82	3.04	2.78	52.
13	14	83	Misc articles of base metal	Home Décor & Hardware	7.68	4.66	3.02	60.
14	15	82	Tools, cutlery	Kitchenware & Tableware	9.61	6.26	3.35	65.
15	16	65	Headgear	Textiles & Apparel	2.34	1.40	0.95	59.
16	17	66	Umbrellas	Textiles & Apparel	0.80	0.40	0.40	49.
17	18	46	Basketware	Seasonal & Festive Goods	1.17	0.71	0.46	60.
18	19	67	Feathers and artificial flowers	Seasonal & Festive Goods	0.39	0.23	0.16	59.

Section 6 — Strategic Conclusions & Recommendations

Key Findings

Based on the Canada Import Gap analysis filtered to the **Yiwu small commodity universe**, the strategic conclusions apply:

1. **High-Priority Segments (large gap, low Chinese penetration)** These represent the imme export opportunity — Canada is buying heavily from other countries, and China's share remains the segment average:

- Categories landing in the **PRIORITY quadrant** (upper-left) should be the first focus for Yiwu strategies
- Look for HS 6 product codes with CN share < 20% AND world market > USD 1 cumulative

2. **Growth Segments (large gap, China already present)** China has demonstrated competitiv but hasn't reached saturation. Opportunity exists to take additional share:

- These require competitive differentiation — price, certification (CSA/UL), or distribution partı
- Useful for suppliers already exporting to Canada seeking volume growth

3. **Yiwu-Specific Advantages** The Yiwu market has structural advantages in:

- **Toys & Games (HS 95)**: deep manufacturing ecosystem, all product types
- **Bags & Leather Goods (HS 42)**: cost leadership, wide SKU variety
- **Home Décor & Hardware (HS 83, 94)**: fast sampling, flexible MOQ
- **Seasonal/Festive (HS 96, 67, 46)**: speed-to-market for trend products

4. **Barriers to Watch**

- Canadian product safety standards (CCPSA, Health Canada) for toys and children's items
- CBSA tariff classifications and anti-dumping measures on certain steel/textile products
- Growing preference for certified, sustainably-sourced products among Canadian retailers

Recommended Next Steps

Priority	Action	Target Se
🔴 High	Identify Canadian retail buyers for top-ranked categories	Toys, Hon
🔴 High	Request HS 6 product samples from Yiwu market matching gap products	All segme
🟡 Medium	Evaluate cost of CSA/UL certification for priority HS 6 lines	Toys, Elec
🟡 Medium	Screen for Canadian distribution partners (wholesalers, Amazon FBA)	Bags, Kitc
🟢 Low	Monitor tariff changes under CUSMA and Canada–China trade relations	All

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Methodology
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In [17]:
```
# ── Export instructions ──────────────────────────────────────────
print('='*65)
print('  EXPORT TO PDF')
print('='*65)
print()
print('Option 1 — Command line (requires nbconvert + LaTeX):')
print('  jupyter nbconvert --to pdf yiwu_canada_opportunity.ipynb')
print()
print('Option 2 — HTML then print to PDF (simpler, recommended):')
print('  jupyter nbconvert --to html yiwu_canada_opportunity.ipynb')
print('  Then open the HTML in your browser and Print → Save as PDF')
print()
print('Option 3 — In Jupyter Lab:')
print('  File → Export Notebook As → PDF')
print()
print('Files generated in this run:')
for f in ['fig_01_kpis.png','fig_02_top25_gaps.png','fig_03_quadrant_all.
          'fig_04_yiwu_segments.png','fig_05_yiwu_quadrant.png',
          'fig_06_segment_detail.png','fig_07_hs6_deepdive.png',
          'fig_08_yiwu_ranking.png','fig_09_trends.png',
          'yiwu_canada_opportunity.xlsx']:
    exists = '✅' if os.path.exists(f) else '⬜'
    print(f'  {exists} {f}')
```

```
=================================================================
  EXPORT TO PDF
=================================================================

Option 1 — Command line (requires nbconvert + LaTeX):
  jupyter nbconvert --to pdf yiwu_canada_opportunity.ipynb

Option 2 — HTML then print to PDF (simpler, recommended):
  jupyter nbconvert --to html yiwu_canada_opportunity.ipynb
  Then open the HTML in your browser and Print → Save as PDF

Option 3 — In Jupyter Lab:
  File → Export Notebook As → PDF

Files generated in this run:
  ✅ fig_01_kpis.png
  ✅ fig_02_top25_gaps.png
  ✅ fig_03_quadrant_all.png
  ✅ fig_04_yiwu_segments.png
  ✅ fig_05_yiwu_quadrant.png
  ✅ fig_06_segment_detail.png
  ✅ fig_07_hs6_deepdive.png
  ✅ fig_08_yiwu_ranking.png
  ✅ fig_09_trends.png
  ✅ yiwu_canada_opportunity.xlsx
```